

9 Review

9.2 Review of Cryptography

Example 9.2.1 Use a shift cipher with key 5 to encrypt the message “The grades are posted”.

ABCDEFGHIJKLMNOPQRSTUVWXYZ
 ABCDEFGHIJKLMNOPQRSTUVWXYZ

Example 9.2.2 Decrypt the following text, using a shift cipher with key 5.

DTZCINICNY

Example 9.2.3 Encrypt the message “You should take the final seriously” using a transposition cipher, and secret key “study”.

Example 9.2.4 The following is an encrypted message using a transposition cipher with secret key “goalie”. What was the message?
ntfrin eafsfm leehat tgiaw

Example 9.2.5 Encrypt the message “The final is on Thursday” using a transposition cipher, and secret key “final”.

Example 9.2.6 The following is an encrypted message using a transposition cipher with secret key “told”. What was the message?
flehen seavclif hnsylgatiirang

Recall the key exchange algorithm

1. Alice and Bob agree publically on values a prime p and generator g .
2. Alice picks some secret number a , while Bob picks some secret number b .
3. Alice computes $A = g^a \bmod p$ and sends it to Bob.
4. Bob computes $B = g^b \bmod p$ and sends it to Alice.
5. Alice computes $B^a \bmod p$, which is $(g^b \bmod p)^a \bmod p$
6. Bob computes $A^b \bmod p$, which is $(g^a \bmod p)^b \bmod p$

Example 9.2.7 Suppose that Alice and Bob publicly choose the prime 17 and generator 3. What does the exchange look like, and where is their secret key?

Example 9.2.8 Your turn! What does the key exchange algorithm look like with $p = 19$ and $g = 13$?